

Activity 3: Capacity, Takt and Bottleneck Analysis

Level: Intermediate Duration: 75 minutes Mapping: K4 A4 Tools: CSV; calculator; worksheet

Goal

Calculate takt, effective capacity and constraint risk, then select an improvement experiment based on facts rather than averages alone.

Learner output

Capacity table, takt comparison, bottleneck decision and countermeasure experiment.

Files in this folder

- README.md / README.pdf - activity guide and detailed procedure
- scenario.md / scenario.pdf - case facts and questions
- worksheet.md / worksheet.pdf - structured response template
- evidence-checklist.md / evidence-checklist.pdf - acceptance record
- data.csv - mock dataset when analysis is required

Detailed procedure

1. Confirm net available time and customer demand from capacity-data.csv.
2. Calculate takt time in minutes per good unit.
3. For each step, calculate rated units per shift from net time and average cycle time.
4. Adjust capacity for planned downtime and first-pass yield.
5. Compare effective capacity with demand and identify the system constraint.
6. Review the 90th-percentile cycle time and explain variation risk.
7. Test one countermeasure scenario by changing only one assumption.
8. Document the decision, expected benefit and verification measure.

Verification

Complete every acceptance item in evidence-checklist.md. Submit the worksheet plus calculations, exported tool output or screenshots named in the worksheet.

Safety, privacy and responsible AI

Use only the supplied scenario data unless your organisation has authorised another dataset. Remove personal, confidential and export-controlled information. AI outputs are suggestions until verified by the named human decision owner.